International Journal of Sport Nutrition and Exercise Metabolism, (Ahead of Print)
https://doi.org/10.1123/ijsnem.2022-0203
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First Published Online: May 24, 2023

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ORIGINAL RESEARCH

Characterizing Hydration Practices in Healthy Young Recreationally Active Adults—Is There Utility in First Morning Urine Sampling?

Colleen X. Muñoz[1] and Michael F. Bergeron[1,2]
[1]University of Hartford, West Hartford, CT, USA; [2]Women's Tennis Association (WTA), St. Petersburg, FL, USA

First morning urine (FMU) assessment would be a practical and convenient solution for clinically acceptable detection of underhydration prior to competition/training, and for the general public. Thus, we thus sought to determine the diagnostic accuracy of FMU as a valid indicator of recent (previous 24 hr, 5 days average) hydration practices. For 5 consecutive days and one final morning, 67 healthy women ($n = 38$) and men ($n = 29$; age: 20 [1] years, body mass index: 25.9 [5.5]) completed 24-hr diet logs for total water intake (from beverages and foods, absolute and relative to body mass), 24-hr urine and FMU collection (last morning only) for osmolality (Osm), specific gravity (SG), and color (Col), and morning blood sampling for plasma osmolality and copeptin. Correlations determined significance and relationship strength among FMU and all other variables. Area under the receiver operating characteristic curves, sensitivity, specificity, and positive likelihood ratios were employed using previously reported values to indicate underhydration (total water intake < 30 ml/kg, osmolality > 500, and >800 mOsm/kg, specific gravity > 1.017, and copeptin > 6.93 pmol/L). FMU_Osm and FMU_SG were significantly correlated ($p < .05$) to all variables except the previous 5-day plasma osmolality. FMU_Col was only significantly correlated with other color time intervals and total water intake per gram. FMU_Osm held greatest utility (area under the receiver operating characteristic curve, sensitivity, and specificity >80%) overall, with the best outcome being FMU_Osm indicating a previous 24-hr osmolality threshold of 500 mOsm/kg (FMU_Osm criterion >710 mOsm/kg and positive likelihood ratio = 5.9). With less effort and cost restriction, FMU is a viable metric to assess underhydration.

***Keywords**:* oral rehydration, osmolality, underhydration, water intake

Reductions in physical performance consistently result with body water deficits incurred during (Cheuvront et al., 2003; James et al., 2019; Judelson et al., 2007; Kraft et al., 2012) and sustained prior to competition (Chapelle et al., 2020; Deshayes et al., 2020), a phenomenon which could be notably compounded with repeated bouts and performance expectations across multiple successive days (Arnaoutis et al., 2018). Beyond physical performance, yet critical to active and nonactive individuals alike, accumulating evidence supports unfavorable health outcomes with regular underhydration (Perrier et al., 2021; Vanhaecke et al., 2020).

Establishing a suitable and practical evidence-informed screening tool aligning hydration practices and performance and health outcomes demands simplicity in low-cost and accurate hydration monitoring. Whereas 24-hr urine samples traditionally appear to be the best available valid assessment of daily hydration practices, spot urine samples (vs. 24-hr collections) are far more convenient and feasible in clinical practice, personal monitoring, and many health-related research scenarios. Moreover, spot urine samples between 1,400 and 2,000 hr were shown to be most closely related to 24-hr assessments in approximating concentration (i.e., urine osmolality [uOsm] and specific gravity [uSG]; Perrier, Demazières, et al., 2013). Although, sampling later in the day could be measurably confounded by dietary solute load and exercise (Bottin et al., 2016; Cheuvront et al., 2015).

Notably, first morning urine (FMU) samples have been effective in differentiating habitual low-volume (LOW) and high-volume (HIGH) drinkers (Hahn, 2023; Perrier, Vergne, et al., 2013). However, whereas FMU is comparatively far less (if at all) detectably impacted by prior dietary solute load and exercise, spot urine samples in general (specific to uOsm and uSG) are not consistently responsive to short-term changes in hydration status (Edwards & Buono, 2022). Nonetheless, Perrier et al. showed that 24-hr urine uOsm in daily LOW was effectively the same as the uOsm of the first morning sample (767 vs. 794 mOsm/kg, respectively). In contrast, 24-hr urine uOsm in the daily HIGH was lower by roughly one third compared with the corresponding FMU sample (371 vs. 590 mOsm/kg, respectively; Perrier, Vergne, et al., 2013). Regardless of a drastic difference in variability between 24 hr and FMU sampling in daily LOW versus HIGH, urine uOsm values generally were above 700 in LOW, whereas HIGH largely remained below 600, leaving some (albeit modest) room for distinguishing these groups using FMU. Moreover, a 24-hr urine collection can be disproportionately influenced by recent fluid intake, whereas an initial morning urine sample appears to better reflect stable corrections of fluid balance and thus denotes a longer-term hydration profile (Hahn, 2021). The utility of FMU and other spot urine samples has been examined in a variety of other ways, including comparative value as a valid indicator of hydration status (Cheuvront et al., 2015; Edwards & Buono, 2022; Muñoz et al., 2013; Perrier, Demazières, et al., 2013; Perrier et al., 2017). However, implications derived from many of these studies, given their respective novel methodologies (e.g., inducing dehydration), inherent control measures (e.g., specific to fluid intake, diet, and physical activity), and relevant primary outcome metrics (e.g., body mass change), are notably challenging to translate and practically apply to hydration indicators responding to free-living daily routines.

Consumers are increasingly aware of and aim to apply personal behaviors and routines that enhance daily hydration. This parallels and is reinforced by greater and broader recognition of the

Bergeron https://orcid.org/0000-0003-1377-0234
Muñoz (cmunoz@hartford.edu) is corresponding author, https://orcid.org/0000-0002-3345-9411

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emerging and developing evidence for maintaining adequate daily hydration as it impacts acute well-being and chronic health outcomes (Armstrong et al., 2020). Therefore, because incorporating a 24-hr urine assessment is widely impractical, an aptly timed spot urine sample would be more viable in determining valid individual hydration status. If FMU assessment is thus contended to be the most practical, valid, and informative solution, reasonable relevant questions that should be posed specifically include (a) what does FMU indicate regarding hydration status (e.g., acute, previous day, or general hydration for the past week), (b) how should an FMU sample potentially be interpreted in a clinical setting, and (c) does FMU have clinical utility related to one's well-being and chronic health?

In this study, we determined the strength of FMU as a valid indicator of recent (previous 24 hr and 5 days average) fluid intake behavior and practices (and thus hydration status) that would otherwise be denoted by evaluating a 5-day dietary intake, 24-hr urine, morning plasma uOsm, and morning circulating copeptin. If validated as such, we then aimed to provide criterion values for practical and clinical utility that could be arguably applicable in the general population. Establishing FMU in this role would have valuable practical implications for readily and objectively determining typical hydration practices and status in individuals that may need to be reinforced or altered.

Methods

This observational study was conducted in accordance with the Declaration of Helsinki and was reviewed and approved by the University Institutional Review Board. The investigation took place during the months of September through February in the Northeastern United States. Participants (67 healthy women [$n = 38$] and men [$n = 29$], who were nonsmokers, exercised <2.5 hr per week, and consumed <500 mg of caffeine per day) first attended a subset grouping (by recruitment/consent order) familiarization visit for measurement of height and weight where they were instructed (with visual and textual examples provided) how to accurately record their dietary intake. These inclusion/exclusion criteria were selected in order to represent habitual hydration practices in the absence of concurrent exercise, which has strong value to exercisers and nonexercisers alike, and to minimize confounding variables. Participants were instructed to save and provide food packaging when possible. Blinding of the study objectives was conducted as practically as possible, by (a) not identifying the intention to primarily examine their total water intake (TWI); (b) informing participants that we simply wanted to observe and that it was important that they do not change typical behaviors related to eating, drinking, and physical activity; and (c) informing them that there would be no judgment whatsoever related to their recorded information.

For 5 consecutive days and one final consecutive morning, within 2 weeks of their respective familiarization visit, each participant completed their own 24-hr diet logs and 24-hr urine collections,, and provided a FMU collection (last morning only; collected prior to arriving at the laboratory) and daily morning blood sampling once they arrived at the laboratory. Participants avoided all alcohol intake for the duration of their participation, completed at least an 8-hr fast overnight (no food or fluids other than water, except no water intake upon awakening), and refrained from exercise for 12 hr each day prior to all experimental morning visits to the laboratory. Upon awakening, participants were instructed to begin urine collection into their current 24-hr collection container. On the morning following Day 5 (last morning), each participant would temporarily halt midstream urination into the 24-hr container used on Day 5, continue urination into a small collection cup for FMU assessment, and then collect any remaining urine into the same 24-hr collection container. FMU samples were collected on the last morning to evaluate the efficacy of using FMU to indicate hydration practices over the previous recent days, which could relieve the time, effort, and cost burden of multiday sampling. Each individual 24-hr urine collection period was started using a new collection container following the participant's respective FMU (i.e., late morning, with the initial urine collection in each new container being their second urination of the respective day).

Self-maintained daily diet logs were brought to the laboratory by each participating individual at every respective subsequent morning visit, reviewed by a trained nutritionist, and additional log-related information was collected via interview as needed. Diet logs were analyzed for TWI (from beverages and foods; Nutritionist Pro, Axxya Systems) absolute and relative to body mass. Urine was measured for uOsm (Touch Micro-OSMETTE, Precision Systems Inc.), uSG (T400 Handheld Refractometer, Reichert), and urine color (uCol; Armstrong et al., 1994). Blood was measured (after separation) for plasma osmolality (pOsm) and plasma copeptin (pCop; Thermo Scientific BRAHMS Copeptin proAVP KRYPTOR). For measurement of pOsm and pCop, blood samples were collected in chilled lithium heparin and dipotassium ethylenediaminetetraacetic acid–treated tubes, respectively, after participants were seated for 5 min. Each blood sample was immediately centrifuged at 2,000*g* for 15 min at 20 °C. For accuracy, pOsm was measured in duplicate or in triplicate, if duplicate variation was >2 mOsm. For pCop measurements, the plasma from each blood sample was briefly centrifuged and then measured in singular per manufacturer's instructions.

All data analyses were conducted using GraphPad Prism version 9.0.0 for Mac (GraphPad Software) with alpha set a priori at .05. For FMU comparison/indication of the previous 5-day hydration practices, an average was calculated for each variable collected over the consecutive 5-day period. Differences in hydration variables among sex and race/ethnicity were evaluated via two-way analysis of variance, variation in hydration variables across days was evaluated via one-way analysis of variance, and Tukey and Sidak (for sex only) post hoc tests were used for all multiple comparisons. Pearson and Spearman (for normally and nonnormally distributed data, respectively) correlations were evaluated for significance and relationship strength among each FMU metric (uOsm, uSG, and uCol) and all other variables. To evaluate diagnostic accuracy, area under the receiver operating characteristic curves (AUC) and positive likelihood ratios were employed for significantly correlated variables using previously reported values to indicate underhydration (TWI < 30 ml/kg [EFSA Panel on Dietetic Products Nutrition and Allergies (NDA), 2010; Institute of Medicine, 2005; Stookey et al., 2017], uOsm > 500 [Perrier et al., 2015] and >800 mOsm/kg [Armstrong et al., 2016], uSG > 1.017 [determined from an average of several investigations due to reported variability; Armstrong et al., 2013, 2019; Johnson et al., 2015; Lemetais et al., 2018; Perrier, Vergne, et al., 2013; Perrier et al., 2017], and pCop > 6.93 pmol/L [Lemetais et al., 2018]). We collected pOsm because it is traditionally still considered by many as a valid hydration status metric; however, given that pOsm does not vary considerably across days regardless of hydration practices (Johnson et al., 2015; Perrier, Vergne, et al., 2013), and

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accordingly, a value to indicate underhydration does not exist, and diagnostic accuracy statistics did not include pOsm. In accordance with established standards (Li & He, 2018), only those variables with an AUC of ≥0.80 were considered valuable indices for the characterization of underhydration over the recent days.

Results

Descriptive statistics of the study participants are shown in Table 1. Participants racially/ethnically identified as (percentage of total) White (45%), Black or African American (26%), Asian (12%), Hispanic or Latino (9%), Black or African American and Hispanic or Latino (3%), Asian and White (3%), and Hispanic or Latino and Native Hawaiian or Pacific Islander (2%). Hydration variables did not differ according to sex or race/ethnicity with the exception of 24-hr TWI (in grams) and 5-day TWI (in grams; Table 2). Figure 1 displays a broad range of habitual TWI and corresponding hydration variables across days and demonstrates minimal daily variation.

Associations Among FMU and Indicators of Recent Hydration Practices

Pearson correlations between (a) FMU_Osm, (b) FMU_SG, and (c) FMU_Col and all other hydration variables at each examined time interval are presented in Table 3. Both FMU_Osm and FMU_SG were significantly correlated to all variables except the previous 5-day pOsm. FMU_Col was only significantly correlated with other uCol time intervals and TWI expressed in grams. Of note, the correlations were weak to (mostly) moderate in strength. Overall, the selected FMU metrics utilized here consistently aligned with, and without any compelling difference between, the previous 5 days and the more acute (i.e., previous 24 hr) hydration index time intervals.

Diagnostic Accuracy of FMU to Indicate Recent Hydration Practices

Of the variables with the highest diagnostic accuracy (Figures 1 and 2, which display only those variables with strongest clinical utility based on an AUC of ≥0.80), the effectiveness of FMU_Osm in detecting an average of 500 mOsm/kg over the previous 24 hr (Figure 2d) was greatest, as evidenced by the highest AUC, sensitivity (SN) and specificity (SP; all of which were greater than the clinically acceptable criterion of 80%). For example, an individual with FMU_Osm of ≥710 mOsm/kg is 5.9 × (as indicated by the positive likelihood ratios) more likely to have a previous 24-hr uOsm of >500 mOsm/kg (Figure 2d). Whether applying an 800 (Figure 2a and 2b) or 500 mOsm/kg threshold (Figure 2c and 2d), the diagnostic accuracy of FMU_Osm was similarly effective in detecting the previous 5-day and 24-hr uOsm, as indicated by AUC, SN, and SP. Whereas the uOsm thresholds differed by 300 mOsm/kg, the respective average (among previous 5-day and 24-hr uOsm) FMU_Osm diagnostic criterion between using 800 or 500 mOsm/kg only differed by 136 mOsm/kg.

FMU_Osm was also successfully utilized (AUC ≥ 0.80) in detecting the previous 5-day uSG and uCol (Figure 2e and 2f, respectively), but not the previous 24-hr uSG (AUC = 0.55, SN = 59, SP = 36) or uCol (AUC = 0.78, SN = 73, SP = 64). However, FMU_Osm was not effective in successfully detecting recent TWI ml·kg·day^{-1} (5-day AUC = 0.79, SN = 77, SP = 73, and 24-hr AUC = 0.77, SN = 72, SP = 73, respectfully), and detection was particularly poor for TWI ml/day (5-day AUC = 0.64, SN = 65, SP = 69, and 24-hr AUC = 0.59, SN = 64, SP = 56, respectfully). In disagreement with the correlative analysis (Table 3), FMU_Osm was not an effective metric for detecting the previous 5 days or concurrent morning's pCop (AUC = 0.67, SN = 60, SP = 62, criterion = 809 mOsm/kg; and AUC = 0.73, SN = 67, SP = 69, criterion = 829 mOsm/kg, respectively).

Whereas the application of an 800 (Figure 3a) versus 500 mOsm/kg (Figure 3b) threshold was similarly effective (AUC ≥0.80) in utilizing FMU_SG to detecting the previous 5-day uOsm, only the 500 mOsm/kg threshold was effective in utilizing FMU_SG to detect the previous 24-hr uOsm (Figure 3c). FMU_SG was also successfully utilized in detecting the previous 5-day uSG (Figure 3d).

Table 1 Descriptive Statistics (Mean [*SD*]) of the Study Participants

Sex	F = 38, M = 29
Age (years)	20 (1)
Height (cm)	169.7 (10.3)
Weight (kg)	74.6 (17.2)
Body fat (%)	24.9 (11.3)
Total exercise (min/week)	96.9 (154.1)

Discussion

We present a unique analysis of FMU as a practical, noninvasive, and cost-effective tool for accurate screening of habitual underhydration, as gauged by two metrics that reflect recent hydration practices and status. Because underhydration can interfere with optimal mood, physical and cognitive health, and wide range of exercise and athletic performance benchmarks, detecting and profiling individual hydration have practical utility in research, clinical care, sports, the military, assorted demanding work/labor scenarios, and free-living individuals interested in fitness and health. Further, previous research has not amply succeeded in providing practical and individually based hydration guidelines for athletes and others to achieve and maintain optional pre-exercise hydration status, which is warranted given the reported high prevalence of underhydration between training/competition bouts (Kostelnik & Valliant, 2023). Our findings confirm the value and support specific utility of FMU_Osm and FMU_SG in representing one's typical recent pattern of hydration over a single day and multiple days. However, FMU_Col does not appear to have similar contributing utility.

FMU has long been appealing because of ease of collection (vs. a 24-hr sample) and in limiting the effects of dietary solute load, intermittent body sweat losses from exercise and/or heat exposure, and consumption of water boluses that could interfere with its valid interpretation (Bottin et al., 2016; Cheuvront et al., 2015). In contrast, spot urine samples collected beyond the first morning void are far more subject to these and other common confounders (e.g., acute fluid intake). Nonetheless, FMU marker criteria and interpretation have remained unclear, in large part due to their study in limited populations (i.e., often athletic and racially undiversified male subjects) and contexts (i.e., experimentally manipulated body water balance). Moreover, there has been minimal comparison with a wide range of accepted biomarkers and

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Table 2 Hydration Variable Mean (*SD*) and Categorical Mean (*SD*) According to Sex and Race/Ethnicity

	All	Women	Men	Hispanic/Latino	Asian	Black/African American	White
FMU_Osm (mOsm/kg)	764 (236)	741 (258)	794 (206)	785 (172)	702 (258)	878 (219)	728 (244)
24-hr uOsm (mOsm/kg)	640 (270)	626 (288)	660 (248)	725 (236)	607 (298)	707 (264)	582 (267)
5-day uOsm (mOsm/kg)	610 (226)	601 (246)	623 (200)	682 (203)	598 (228)	696 (206)	542 (272)
FMU_SG	1.021 (0.006)	1.021 (0.007)	1.021 (0.005)	1.024 (0.003)	1.019 (0.006)	1.022 (0.006)	1.020 (0.006)
24-hr uSG	1.016 (0.007)	1.017 (0.006)	1.016 (0.007)	1.014 (0.005)	1.019 (0.006)	1.016 (0.007)	1.017 (0.006)
5-day uSG	1.017 (0.006)	1.016 (0.006)	1.017 (0.005)	1.018 (0.005)	1.016 (0.006)	1.018 (0.005)	1.015 (0.006)
FMU_Col	4 (1)	5 (1)	4 (1)	5 (1)	4 (1)	4 (1)	4 (2)
24-hr uCol	4 (2)	4 (2)	3 (2)	4 (1)	3 (2)	4 (2)	3 (2)
5-day uCol	4 (1)	4 (1)	3 (1)	4 (1)	4 (1)	4 (1)	3 (1)
Concurrent morning pCop (pmol/L)	7.5 (4.7)	6.6 (4.3)	8.6 (5.0)	8.9 (5.0)	8.8 (7.8)	6.8 (2.6)	7.0 (4.4)
5-day pCop (pmol/L)	8.5 (4.9)	7.6 (4.3)	9.7 (5.3)	10.2 (7.0)	10.1 (8.4)	7.9 (2.9)	7.8 (3.1)
Concurrent morning pOsm (mOsm/kg)	291 (5)	290 (5)	293 (5)	294 (6)	290 (6)	291 (6)	291 (5)
5-day pOsm (mOsm/kg)	292 (4)	290 (4)	294 (3)	293 (4)	290 (5)	291 (5)	291 (4)
24-hr TWI (g)	2,422 (1,205)	2,135 (1,076)	2,798 (1,278)*	2,041 (775)	2,918 (1,901)[†,‡]	2,076 (790)	2,580 (1,193)[†,‡]
5-day TWI (g)	2,581 (1,094)	2,285 (1,051)	2,968 (1,044)*	2,256 (676)	2,797 (1,421)[†,‡]	2,123 (733)	2,877 (1,181)[†,‡]
24-hr TWI (ml/kg)	34 (19)	33 (19)	35 (19)	25 (9)	51 (34)	25 (11)	36 (15)
5-day TWI (ml/kg)	36 (17)	36 (18)	37 (15)	28 (9)	48 (25)	26 (11)	40 (14)

Note. FMU = first morning urine; Osm = osmolality; SG = specific gravity; Col = color; TWI = total water intake; pOsm = plasma osmolality; pCop = plasma copeptin; uOsm = urine osmolality; uSG = urine specific gravity; uCol = urine color.
*Differences from women, [†]from Hispanic/Latino, and [‡]from Black/African American ($p < .05$).

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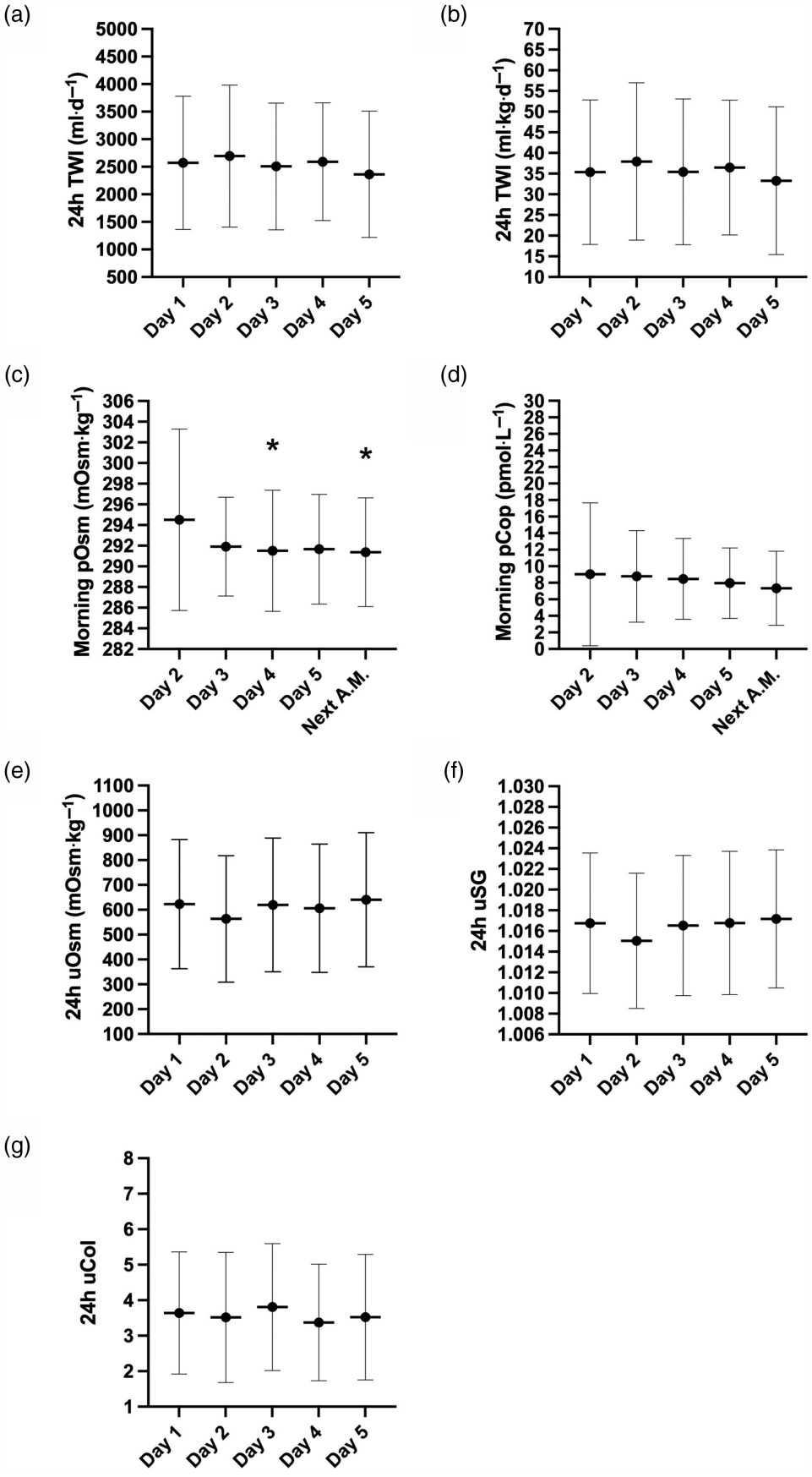

Figure 1 — Habitual TWI and corresponding hydration variables across days. Panels a and b represent 24-hr TWI in absolute and relative to body mass units, respectively. Panels c and d represent morning blood measurements of pOsm and pCop, respectively. Panels e–g represent 24 urine measurements of uOsm, uSG, and uCol, respectively. *Differences from Day 2 ($p < .05$). TWI = total water intake; pOsm = plasma osmolality; pCop = plasma copeptin; uOsm = urine osmolality; uSG = urine specific gravity; uCol = urine color.

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Table 3 Correlations Between FMU (Osm, SG, and Col) and All Other Hydration Variables at Each Examined Time Interval

	FMU_Osm			FMU_SG			FMU_Col		
	p	*r*	*n*	*p*	*r*	*n*	*p*	*r*	*n*
24-hr uOsm	<.001	.783*	58	<.001	.698*	62	.174	.171	65
5-day uOsm	<.001	.709*	59	<.001	.671*	64	.089	.209	67
24-hr uSG	<.001	.529*	59	<.001	.463*	64	.127	.189	67
5-day uSG	<.001	.671*	59	<.001	.631*	64	.088	.210	67
24-hr uCol	<.001	.604*	59	<.001	.562*	64	<.001	.485*	67
5-day uCol	<.001	.609*	59	<.001	.556*	64	.001	.396*	67
Concurrent morning pCop	.002	.256*	57	.001	.403*	62	.349	.079	65
5-day pCop	.011	.328*	59	.022	.285*	64	.331	.074	67
Concurrent morning pOsm	.004	.381*	56	.002	.386*	61	.390	.109	64
5-day pOsm	.521	.085	59	.533	.079	64	.813	−.029	67
24-hr TWI (g)	.013	−.322*	59	.006	−.337*	64	.043	−.247*	67
5-day TWI (g)	.001	−.427*	59	<.001	−.440*	64	.018	−.296*	67
24-hr TWI (ml/kg)	<.001	−.472*	59	<.001	−.437*	64	.264	−.138	67
5-day TWI (ml/kg)	<.001	−.580*	59	<.001	−.548*	64	.086	−.211	67

Note. Variable sample sizes were attributable to missing samples or analytical error. TWI = total water intake; pOsm = plasma osmolality; pCop = plasma copeptin; uOsm = urine osmolality; uSG = urine specific gravity; uCol = urine color; FMU = first morning urine; Osm = osmolality; SG = specific gravity; Col = color.
*Statistically significant correlation ($p < .05$).

conventional time points, and diagnostic accuracy statistics have not been widely employed.

uOsm of the urine has been noted as being more indicative of overall osmotic balance and renal function than uSG and uCol (Jacobson et al., 1962; Kamel et al., 1990). This corresponds with FMU_Col failing to reach and FMU_SG inconsistently (compared to FMU_Osm) achieving acceptable clinical utility limits as defined in this investigation (AUC ≥ 0.80); although, FMU_SG often approached acceptability and should thus be investigated further. In this study, however, we specifically demonstrated the valid clinical utility of FMU_Osm in detecting underhydration according to urinary variables in particular. While many consider elevated arginine vasopressin (and therefore pCop) and low TWI as two additional components indicating underhydration, we suspect inadequate diagnostic accuracy of FMU with pCop and TWI could be functions of sample timing (morning pCop) and self-reporting (TWI). These considerations accordingly warrant investigation with variations of the methodology presented here. The practical strength and scope of translation of this finding are supported by the determined efficacy in applying established diagnostic accuracy statistics for detection of 24-hr uOsm using a 500 mOsm/kg threshold in healthy young female and male individuals from diversified ethnic and racial backgrounds going about their routine daily lives. Moreover, we provide criterion values for specific comparison of the past 24 hr or 5 days and of several accepted hydration biomarkers that can be used with confidence to screen for underhydration in individuals without known (i.e., kidney disease) or anticipated (i.e., advanced age-related) renal impairments.

Previously, researchers and clinical practitioners lacked clear and corroborated criteria for screening underhydration with FMU that correspond with more accepted (albeit invasive) hydration status biomarkers. This left an unacceptable option to contrast individual serial measurements or compare and interpret values to research with deliberate and specified heat- and exercise-induced dehydration procedures and findings that may be decidedly inappropriate in extrapolating to an individual hydration assessment at-hand. By only adopting those variables with an AUC of ≥0.80, clinical distinction of underhydration or not becomes clearer and standardized versus simply adopting mean values from investigations with dissimilar methodologies and scenarios.

In the absence of diagnostic accuracy statistics, Perrier, Vergne, et al. (2013) reported mean FMU_Osm in LOW and HIGH (defined as <1 L/day and >2.5 L/day, respectively) of 794 versus 580 mOsm/L, respectively, and FMU_SG of 1.021 versus 1.016, respectively. Comparatively, our data generated only two variables with acceptable diagnostic accuracy, which nearly align with the indication of LOW from Perrier et al. With confirmation and increasing substantiation of a 500 versus 800 mOsm/kg uOsm threshold for determining adequate TWI (Perrier et al., 2015; Stookey et al., 2020), we examined both thresholds in this investigation. We aimed to contribute to further understanding and establishing agreement that the 500 versus 800 mOsm/kg threshold generally produced as good if not better diagnostic accuracy outcomes. The converging diagnostic criteria observed between the 500 versus 800 mOsm/kg uOsm thresholds are arguably due to the recent (e.g., overnight) greater presence and action of arginine vasopressin compared with spot samples collected later in the day. Whereas FMU_SG nearly reached clinical acceptability, our data suggest using an FMU_SG criterion of 1.021 over the past 24 hr and 5 days with a uOsm threshold of 500 mOsm/L, which mimics the results of Perrier, Vergne, et al. (2013).

Cheuvront et al. (2010) and Muñoz et al. (2013) have also applied diagnostic accuracy statistics to hydration biomarkers, although these methodologies incorporated use of deliberate and specified exercise and heat exposure to enhance body water losses and for application to athletic, military, and labor personnel. Accordingly, Cheuvront et al. proposed an FMU_Osm and FMU_SG criteria for dehydration of 831 mOsm/L and 1.025, respectively. Muñoz et al. examined spot urine samples in those who underwent heat and exercise exposure at each percentage body mass loss, after implementing adequate controls such as matching food and beverage consumption 24 hr prior and providing a

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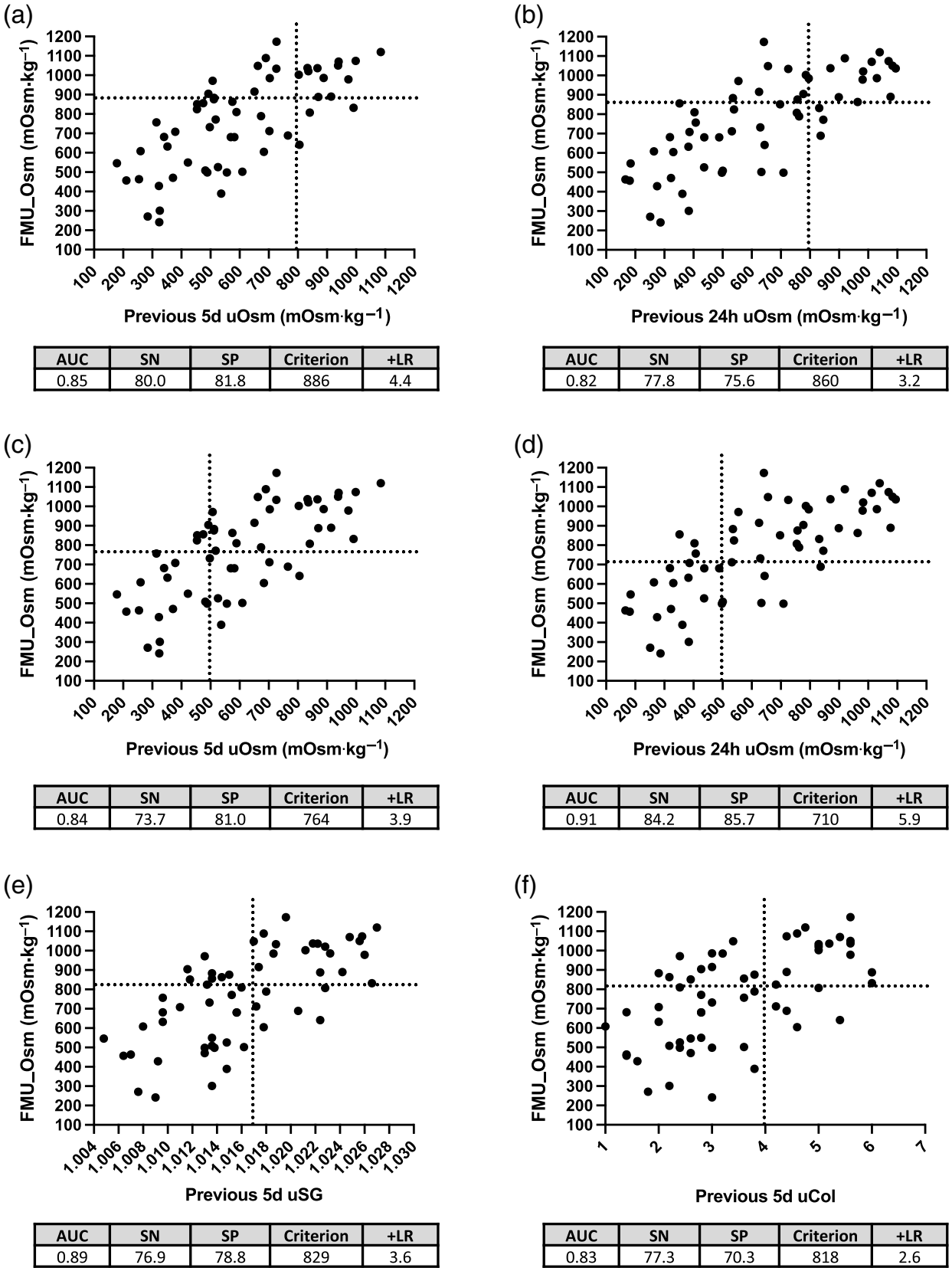

AUC	SN	SP	Criterion	+LR
0.85	80.0	81.8	886	4.4

AUC	SN	SP	Criterion	+LR
0.82	77.8	75.6	860	3.2

AUC	SN	SP	Criterion	+LR
0.84	73.7	81.0	764	3.9

AUC	SN	SP	Criterion	+LR
0.91	84.2	85.7	710	5.9

AUC	SN	SP	Criterion	+LR
0.89	76.9	78.8	829	3.6

AUC	SN	SP	Criterion	+LR
0.83	77.3	70.3	818	2.6

Figure 2 — Acceptable diagnostic accuracy (variables with strongest clinical utility based on an AUC of ≥0.80) with FMU_Osm at selected time intervals, with corresponding AUC, SN, SP, respective classification criteria, and +LR. Panels a and b represent an 800 mOsm/kg uOsm threshold. Panels c and d represent a 500 mOsm/kg uOsm threshold. Panel e represents the 1.017 uSG threshold. Panel f represents the 4 uCol threshold. AUC = area under the receiver operating characteristic curve; SN = sensitivity; SP = specificity; uOsm = urine osmolality; uSG = urine specific gravity; uCol = urine color; +LR = positive likelihood ratios.

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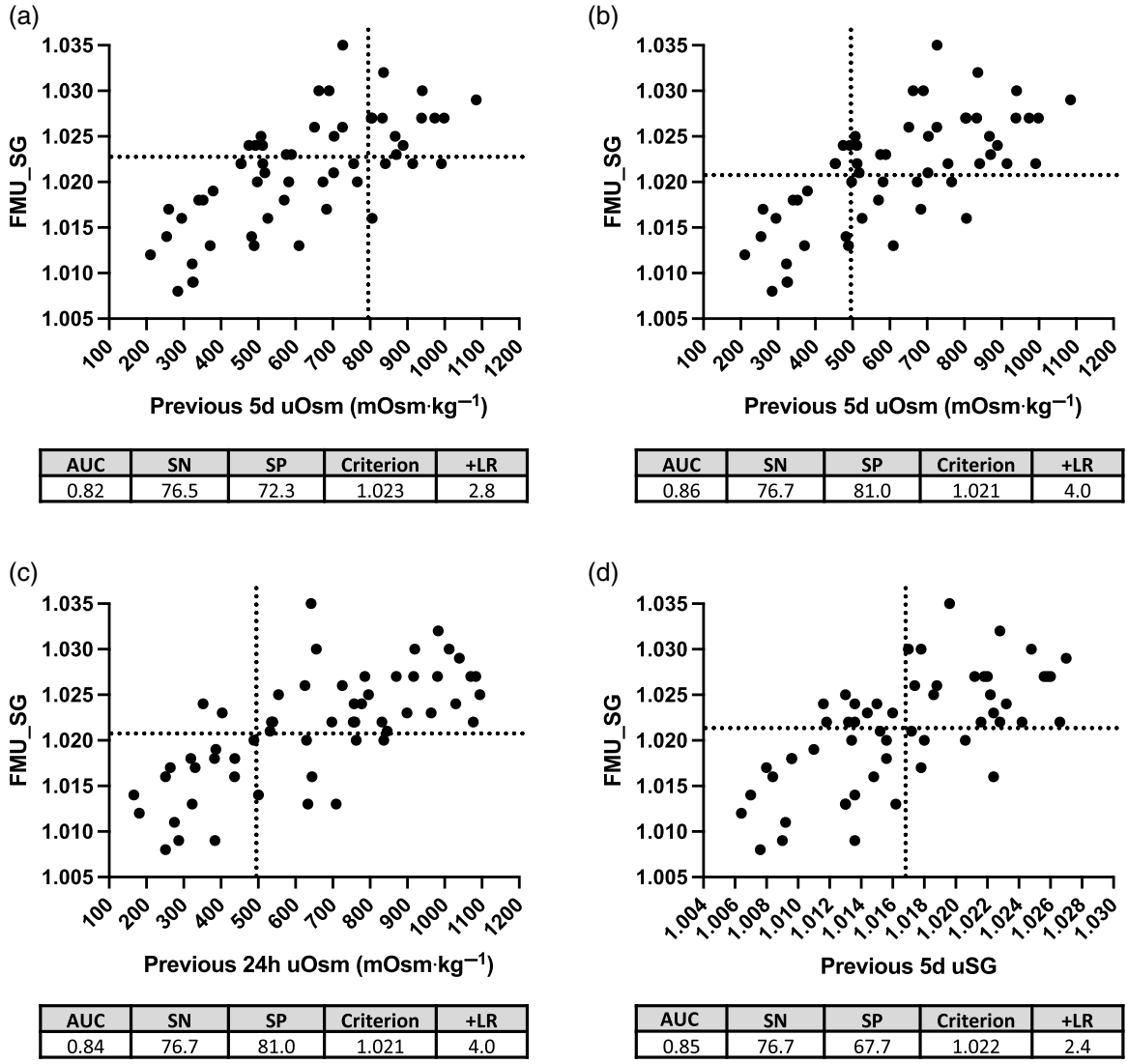

Figure 3 — Acceptable diagnostic accuracy (variables with strongest clinical utility based on an AUC of ≥0.80) with FMU_SG at selected time intervals, with corresponding AUC, SN, SP, respective classification criteria, and +LR. Panel a represents an 800 mOsm/kg uOsm threshold. Panels b and c represent a 500 mOsm/kg uOsm threshold. Panel d represents the 1.017 uSG threshold. AUC = area under the receiver operating characteristic curve; SN = sensitivity; SP = specificity; uOsm = urine osmolality; uSG = urine specific gravity; uCol = urine color; +LR = positive likelihood ratios.

standardized breakfast the morning of each experimental session. This investigation proposed FMU_Osm and FMU_SG dehydration criteria of 631 mOsm/L and 1.020. FMU sampling consistently produces higher values than sampling times later in the day, as does exercise and heat exposure, which explains the variability in Cheuvront and Muñoz criteria for dehydration versus those thresholds proposed by Perrier and the current investigation for those going about their daily lives. Coinciding with the generalizability of our findings to a broader population, 13 (~20%) of the 67 participants self-reported exceeded 150 min (2.5 hr) of exercise per week. While this proportion of participants did not align with the intended amount of exercise exposure, we believe this phenomenon only strengthens the generalizability of our findings, considering that ~22%–23% of the population regularly achieves exercise recommendations. Further, the lack of agreement between FMU and pOsm aligned with previous findings and interpretations pertaining to tight regulation of pOsm often at the expense of other water compartments and, therefore, does not vary substantially in accordance with hydration practices in the absence of exercise.

Devices intended for individual/personal hydration assessment should strongly consider instructing users to sample with FMU to avoid daily behavioral confounders. Whereas we have provided criterion values for this purpose, they should be validated on a larger scale. These findings and implications could also be useful for researchers looking to identify hydration status for trial eligibility and for athletic performance and clinical monitoring. With less effort and cost restriction, FMU is a viable metric to assess routine hydration practices. With further well-designed broadly inclusive studies across regions, ages, and comorbidities, the potential clinical and personal applications could extend to other (and larger) populations in mitigating related risk of chronic illness and general well-being and performance.

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Acknowledgments

The authors would like to sincerely thank the participants who devoted their time and effort to this investigation and the many research assistants for their dedication to this work. Funding was provided by the Drinking Water Research Foundation and the University of Hartford. **Author Contribution:** *Conceptualization, funding acquisition, methodology, writing, reviewing, and editing of the original draft*: Muñoz, Bergeron. *Data curation, formal analysis, and project administration*: Muñoz. *Final version of the paper*: Muñoz, Bergeron.

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